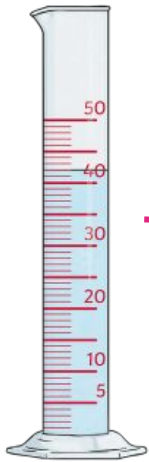




Disappearing Ink

Method:

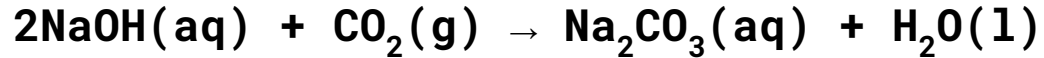
1. Place 10 drops of **ethanol** in a small beaker.
2. Add a 5 drops of **PP indicator** solution.
3. Add 2 drops of **NaOH** solution -
dropwise, to produce a **deep pink** colour in the solution.
4. Using a small paint brush, test the 'disappearing ink' on a white page.



How does it work?

Sodium hydroxide reacts with carbon dioxide in the air.

This forms → Sodium carbonate.



How does it work?

Sodium carbonate is **less basic** than sodium hydroxide.

- Sodium carbonate pH < Sodium hydroxide pH.

This causes the indicators to change colour.



The alcohol evaporates and leaves a clear and colourless residue.

Q&A

Questions

The colour change occurs because sodium hydroxide reacts with a gas in the air.

1. Which gas in the air causes this colour change?
2. Write a word equation for the reaction.
3. Write a formula equation for the reaction.

Answers

1. Carbon dioxide
2. Sodium hydroxide + carbon dioxide → sodium carbonate + water
3. $2\text{NaOH} + \text{CO}_2 \rightarrow \text{Na}_2\text{CO}_3 + \text{H}_2\text{O}$